(Please write your Enrollment Number) Enrollment No. 18001172023

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B. Tech. (CSEAI), Semester: 6th
(May, 2026)

Subject Code: BAI 322	Subject: Data Warehousing & Business Intelligence
Time :3 Hours	Maximum Marks: 60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(4*5=20)
a.	Identify whether the following systems are DBMS or Data Warehouse, and justify: (i) Supports frequent updates and transactions (ii) Stores historical, time-variant data (iii) Optimized for query and analysis (iv) Used for day-to-day operations	4
b.	Identify the following as Fact or Dimension, and justify: (i) Sales Amount (ii) Time (iii) Customer (iv) Quantity Sold	4
c.	Explain the concept of Dimensional Data Modelling. Explain how Data Cube support multidimensional analysis.	4
d.	Given the following dataset: Point | Feature X | Feature Y | Class A | 2 | 3 | Class 1 B | 3 | 4 | Class 1 C | 6 | 7 | Class 2 D | 7 | 8 | Class 2 Classify the point **(4,5)** using **KNN (k = 3)** with Euclidean distance. Show all steps and determine the final class.	4
e.	Given the following transactions: T1: {Milk, Bread} T2: {Milk, Bread, Butter} T3: {Bread, Butter} T4: {Milk, Bread} Calculate using Apriori Algorithm: (i) Support of {Milk, Bread} (ii) Confidence of rule Milk → Bread (iii) Calculate the final rules with support value 60% and confidence 80%.	4

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	UNIT I		CO Mapping
Q2	Describe the three-tier architecture of a data warehouse. Explain the role of each tier and how this architecture supports efficient data analysis.	(10)	CO1
Q3	An organization integrates data from multiple operational systems into a centralized repository. The data is first loaded into the system and then transformed within the database engine to support large-scale processing. Identify the approach used and explain its working and advantages.	(10)	CO1
	UNIT - II		CO Mapping
Q4	Explain and compare the following schemas: • Star Schema • Snowflake Schema • Fact Constellation Schema Discuss their advantages and limitations.	(10)	CO2
Q5	Explain the following OLAP operations with examples: • Drill-down • Roll-up • Slice and Dice • Pivot	(10)	CO2
	UNIT III		CO Mapping
Q6	Explain Data Mining in detail. Discuss the role of KDD process in business decision-making and various application areas. Explain each phase and its significance in extracting useful knowledge.	(10)	CO3
Q7	Explain the concept of Classification and Prediction with suitable examples. Discuss the Decision Tree approach in detail.	(10)	CO3
	UNIT IV		CO Mapping
Q8	Explain the K-means algorithm in detail. Given the following data points: (2, 2), (3, 3), (8, 8), (9, 9) Apply **K-means clustering (k = 2)** with initial centroids as (2,2) and (8,8). Find the final clusters.	(10)	CO4
Q9	Explain the following: • Neural Networks and Genetic Algorithms • Agglomerative and Divisive Clustering	(10)	CO4

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